0.1 Riemann-Stieltjes Integral

Definition 0.1.1. A finite subset $P \subset [a,b]$ containing a, b is called a *Partition* of $[a,b]$.
Conventionally, denote $P = \{x_0, x_1, \ldots, x_n\}$ where $a = x_0 < x_1 < \cdots < x_n = b$.
The set of Partitions of $[a,b]$ is denoted as $\mathcal{P}[a,b]$.

Definition 0.1.2. Let $f : [a,b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a,b] \to \mathbb{R}$ be a monotone-increasing.
Let a partition $P \in \mathcal{P}[a,b]$ be given, and denote $P = \{x_0, x_1, \ldots, x_n\}$. For each $1 \le i \le n$, define

$$\Delta\alpha_i \overset{\text{def}}{=} \alpha(x_i) - \alpha(x_{i-1}) \underset{\alpha \text{ mono.}}{\geq} 0$$

and

$$M_i \overset{\text{def}}{=} \sup\{f(t) \mid x_{i-1} \le t \le x_i\} \text{ and } m_i \overset{\text{def}}{=} \inf\{f(t) \mid x_{i-1} \le t \le x_i\}$$

And, define *Upper Sum* and *Lower Sum* as:

$$U(f,P,\alpha) \overset{\text{def}}{=} \sum_{i=1}^{n} M_i \cdot \Delta\alpha_i \text{ and } L(f,P,\alpha) \overset{\text{def}}{=} \sum_{i=1}^{n} m_i \cdot \Delta\alpha_i$$

Now, define *Upper integral* and *Lower integral* as:

$$\overline{\int_a^b} f d\alpha \overset{\text{def}}{=} \inf_{P \in \mathcal{P}[a,b]} U(f,P,\alpha) \text{ and } \underline{\int_a^b} f d\alpha \overset{\text{def}}{=} \sup_{P \in \mathcal{P}[a,b]} L(f,P,\alpha)$$

We say that f is *Riemann-Stieltjes Integrable* with respect to α if:

$$\overline{\int_a^b} f d\alpha = \underline{\int_a^b} f d\alpha$$

The set of Riemann-Stieltjes Integrable functions with respect to α as $\mathcal{R}(\alpha)$.

Lemma 0.1.3. Let $f : [a,b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a,b] \to \mathbb{R}$ be a monotone-increasing.
For any partitions $P, Q \subset [a,b]$, and a refinement $P^* \supset P$,

$$L(f,P,\alpha) \le L(f,P^*,\alpha) \le \underline{\int_a^b} f d\alpha \le \overline{\int_a^b} f d\alpha \le U(f,P^*,\alpha) \le U(f,P,\alpha)$$

and

$$L(f,P,\alpha) \le U(f,Q,\alpha)$$

Proof of Lemma 0.1.3.
Let $P^* \supseteq P$ be a refinement of P. Since Partition contains only finite elements,
WLOG, assume that $P^* = P \cup \{s\}$ s.t. $s \notin P = \{x_0, \ldots, x_n\}$
Then, for some $1 \le j \le n$ $x_{j-1} < s < x_j$. Now,

$$\begin{aligned} U(f,P,\alpha) - U(f,P^*,\alpha) &= \sup_{t \in [x_{j-1}, x_j]} f(t) \cdot [\alpha(x_j) - \alpha(s)] + \sup_{t \in [x_{j-1}, x_j]} f(t) \cdot [\alpha(s) - \alpha(x_{j-1})] \\ &\quad - \sup_{t \in [x_{j-1}, s]} f(t) \cdot [\alpha(x_j) - \alpha(s)] - \sup_{t \in [s, x_j]} f(t) \cdot [\alpha(s) - \alpha(x_{j-1})] \\ &\ge 0. \end{aligned}$$

$\blacksquare$

0.1.1 Properties of Riemann-Stieltjes Integral

Theorem 0.1.4. Let $f : [a,b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a,b] \to \mathbb{R}$ be monotone-increasing. TFAE:

1. f is Riemann-Stieltjes Integrable with respect to α on $[a,b]$.
2. For any $\varepsilon > 0$, there exists a Partition $P \in \mathcal{P}[a,b]$ such that $U(f,P,\alpha) - L(f,P,\alpha) < \varepsilon$.
3. For some $A \in \mathbb{R}$, for any $\varepsilon > 0$, there exists $P_0 \in \mathcal{P}[a,b]$ such that

$$\forall P \in \mathcal{P}[a,b],\ \ P \supseteq P_0 \implies |S(f,P,\alpha) - A| < \varepsilon$$

where Riemann-Stieltjes Sum $S(f,P,\alpha) \overset{\text{def}}{=} \sum_{i=1}^{n} f(t_i)[\alpha(x_i) - \alpha(x_{i-1})]$, for some $t_i \in [x_{i-1}, x_i]$.

Lemma 0.1.5. Let $f : [a,b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a,b] \to \mathbb{R}$ be a monotone-increasing.
If $f, g \in \mathcal{R}(\alpha)$ and $c \in \mathbb{R}$, then $f + g,\ \ cf \in \mathcal{R}(\alpha)$ with

$$\int_a^b (f+g)d\alpha = \int_a^b f d\alpha + \int_a^b g d\alpha$$
$$\int_a^b (cf)d\alpha = c\int_a^b f d\alpha$$

Lemma 0.1.6. Let $f : [a,b] \to \mathbb{R}$ be a bounded map, and $\alpha : [a,b] \to \mathbb{R}$ be a monotone-increasing, and $a < c < b$.

f is Stieltjes Integrable on $[a,b]$ *if and only if* f is Stieltjes Integrable on $[a,c]$ and $[c,b]$

In this case, the following equality holds:

$$\int_a^b f d\alpha = \int_a^c f d\alpha + \int_c^b f d\alpha$$

Lemma 0.1.7. Let $f : [a,b] \to \mathbb{R}$ be a bounded map, and $\alpha, \beta : [a,b] \to \mathbb{R}$ be monotone-increasing.
If $f \in \mathcal{R}(\alpha)$, $f \in \mathcal{R}(\beta)$ and $c > 0$, then $f \in \mathcal{R}(\alpha + \beta)$ and $f \in \mathcal{R}(c\alpha)$ and $|f| \in \mathcal{R}(\alpha)$ with

$$\int_a^b f d(\alpha+\beta) = \int_a^b f d\alpha + \int_a^b f d\beta$$
$$\int_a^b f d(c\alpha) = c\int_a^b f d\alpha$$
$$\left|\int_a^b f d\alpha\right| \le \int_a^b |f| d\alpha$$

Proof of Thm 0.1.4

1) ⇒ 2). Since $f \in \mathcal{R}(\alpha)$, given $\varepsilon > 0$, there exists a Partition $P \in \mathcal{P}[a,b]$ s.t

$$\int_a^b f\, d\alpha \le \mathcal{U}(f,P,\alpha) < \int_a^b f\, d\alpha + \frac{\varepsilon}{2} \quad \text{and} \quad \int_a^b f\, d\alpha \ge \mathcal{L}(f,P,\alpha) > \int_a^b f\, d\alpha - \frac{\varepsilon}{2}$$

Ask the next step to a toddler on the street.

2) ⇒ 1). It is clear, because for any partition $P \in \mathcal{P}[a,b]$,

$$\overline{\int_a^b} f\, d\alpha - \underline{\int_a^b} f\, d\alpha \le \mathcal{U}(f,P,\alpha) - \mathcal{L}(f,P,\alpha)$$

To show 1) ⇔ 3), we prove a statement: If 2) holds, then for any $1 \le i \le n$,

for arbitrary $s_i, t_i \in [x_{i-1}, x_i]$, $\sum_{i=1}^{n} |f(s_i) - f(t_i)| \cdot \Delta\alpha_i < \varepsilon$.

Since $|f(s_i) - f(t_i)| \le M_i - m_i$,

$$\sum_{i=1}^{n} |f(s_i) - f(t_i)| \cdot \Delta\alpha_i \le \mathcal{U}(f,P,\alpha) - \mathcal{L}(f,P,\alpha) < \varepsilon.$$

1) ⇒ 3). Suppose $f \in \mathcal{R}(\alpha)$. Then, by 2), there exists a Partition $P_0 \in \mathcal{P}[a,b]$ s.t

$$\mathcal{U}(f,P_0,\alpha) - \mathcal{L}(f,P_0,\alpha) < \varepsilon$$

Now, for any refinement $P \supset P_0$,

$$\mathcal{U}(f,P,\alpha) - \mathcal{L}(f,P,\alpha) \le \mathcal{U}(f,P_0,\alpha) - \mathcal{L}(f,P_0,\alpha) < \varepsilon$$

Clearly, regardless of choice of t_i,

$$\mathcal{L}(f,P,\alpha) \le S(f,P,\alpha) \le \mathcal{U}(f,P,\alpha)$$

and

$$\mathcal{U}(f,P,\alpha) \ge \int f\, d\alpha > \mathcal{L}(f,P,\alpha)$$

Therefore, $\left| S(f,P,\alpha) - \int f\, d\alpha \right| \le \mathcal{U}(f,P,\alpha) - \mathcal{L}(f,P,\alpha) < \varepsilon$.

3) ⇒ 1) is trivial.

「Proof of lemma 0.1.5.

Let $f, g \in \mathcal{R}(\alpha)$ be given. Given $\varepsilon > 0$, there exists a partition $P \in \mathcal{P}[a,b]$ such that

$$U(f,P,\alpha) - L(f,P,\alpha) < \frac{\varepsilon}{2} \text{ and } U(g,P,\alpha) - L(g,P,\alpha) < \frac{\varepsilon}{2}$$

(Technically, take $P_1, P_2 \in \mathcal{P}[a,b]$ for f, g respectively, and put $P = P_1 \cup P_2$) Now,

$$U(f+g,P,\alpha) - L(f+g,P,\alpha) \leq U(f,P,\alpha) + U(g,P,\alpha) - L(f,P,\alpha) - L(g,P,\alpha) < \varepsilon.$$

And the equality is obtained by: given $\varepsilon > 0$, there exists a partition $P \in \mathcal{P}[a,b]$ s.t

$$\begin{aligned}\int_a^b f\,d\alpha + \int_a^b g\,d\alpha - \varepsilon &\leq L(f,P,\alpha) + L(g,P,\alpha) \leq L(f+g,P,\alpha) \\ &\leq \int_a^b f+g\,d\alpha \\ &\leq U(f+g,P,\alpha) \leq U(f,P,\alpha) + U(g,P,\alpha) \leq \int_a^b f\,d\alpha + \int_a^b g\,d\alpha + \varepsilon\end{aligned}$$

「Proof of lemma 0.1.6.

Suppose that $f \in \mathcal{R}(\alpha)$ on $[a,b]$. Given $\varepsilon > 0$, there exists $P \in \mathcal{P}[a,b]$ s.t

$$U(f,P,\alpha) - L(f,P,\alpha) < \varepsilon.$$

Put $P^* = P \cup \{c\}$. Then, for $P_1 = P^* \cap [a,c]$ and $P_2 = P^* \cap [c,b]$. Now,

$$\begin{aligned}\overline{\int_a^c} f\,d\alpha - \underline{\int_a^c} f\,d\alpha + \overline{\int_c^b} f\,d\alpha - \underline{\int_c^b} f\,d\alpha &\leq U(f,P_1,\alpha) - L(f,P_1,\alpha) + U(f,P_2,\alpha) - L(f,P_2,\alpha) \\ &= U(f,P^*,\alpha) - L(f,P^*,\alpha) < \varepsilon.\end{aligned}$$

Similarly, the Converse holds.

Thm. Let $f:[a,b]\to\mathbb{R}$ be a bounded map and continuous except finitely many points. Then f is Riemann integrable.

Proof. Define $P_n=[a,b]\cap\left\{\frac{k}{2^n}\,\middle|\,k\in\mathbb{Z}\right\}$. Let A be a set of discontinuous point of f.

Put N be a number of elements of A. Now, for any $n\in\mathbb{N}$, there is at most $2N$ for $i\in\mathbb{N}$

s.t. $A\cap[x_{i-1},x_i]\neq\emptyset$

where $x_i\in P_n$. Given $\varepsilon>0$, take $n\in\mathbb{N}$ s.t. $2\cdot\sup|f|\cdot 2N\cdot\frac{1}{2^n}<\frac{\varepsilon}{2}$.

Meanwhile, f is continuous on the set

$$C=\bigcup_{A\cap[x_{i-1},x_i]=\emptyset}[x_{i-1},x_i]$$

Therefore there is a partition P' on C s.t. $\mathcal{U}(f,P')-\mathcal{L}(f,P')<\frac{\varepsilon}{2}$.

Finally, for $P=P_n\cup P'$,

$\mathcal{U}(f,P)-\mathcal{L}(f,P)<\frac{\varepsilon}{2}+\frac{\varepsilon}{2}<\varepsilon.$ ∎

0.1.2 Riemann-Stieltjes Integrable

Theorem 0.1.8. If $f : [a, b] \to \mathbb{R}$ is continuous, then $f \in \mathcal{R}(\alpha)$ for any monotone increasing $\alpha : [a, b] \to \mathbb{R}$.

Theorem 0.1.9. If $f : [a, b] \to \mathbb{R}$ is monotone and $\alpha : [a, b] \to \mathbb{R}$ is monotone increasing continuous, then $f \in \mathcal{R}(\alpha)$.

Theorem 0.1.10. If $f : [a, b] \to \mathbb{R}$ is map with $m \le f(t) \le M$ for all $t \in [a, b]$ and ϕ is continuous on $[m, M]$, then $\phi \circ f$ is Riemann-Stieltjes integrable with respect to α on $[a, b]$

Theorem 0.1.11. If $\alpha : [a, b] \to \mathbb{R}$ is continuously differentiable monotone increasing, then

$$f \in \mathcal{R}(\alpha) \text{ if and only if } f\alpha' \in \mathcal{R}(x)$$

In this case,

$$\int_a^b f(x) d\alpha(x) = \int_a^b f(x)\alpha'(x) dx$$

Proof of Thm 0.1.8.

Given $\varepsilon > 0$, take $\eta > 0$ s.t $[\alpha(b) - \alpha(a)] \cdot \eta < \varepsilon$.

Since f is continuous on the compact set $[a,b]$, it is continuous uniformly.

Thus, there exists $\delta > 0$ s.t

$$\forall x, y \in [a,b], \quad |x-y| < \delta \Rightarrow |f(x) - f(y)| < \eta.$$

Now, let $P \in \mathcal{P}[a,b]$ be a partition such that for all i, $\Delta x_i < \delta$ ($P = \{x_0, \cdots, x_n\}$)

Then, for all i, $M_i - m_i \le \eta$. Now,

$$U(f, P, \alpha) - L(f, P, \alpha) = \sum_{i=1}^n (M_i - m_i) \cdot \Delta\alpha_i \le \sum_{i=1}^n \eta \cdot \Delta\alpha_i = \eta \cdot \sum_{i=1}^n \Delta\alpha_i = \eta \cdot [\alpha(b) - \alpha(a)] < \varepsilon.$$

「Proof of Thm 0.1.9. Alternatively, since monotone function can have only countively simple discontinuity if no such shared discontinuities with f and α, then countively discontinuous f is R.S. integrable. But, in this, prove by elementary, using properties of monotone.

Since α is continuous, for each $n \in \mathbb{N}$, there exists a partition $P_n \in \mathcal{P}[a,b]$,

$$\Delta\alpha_i = \frac{\alpha(b)-\alpha(a)}{n} \left(\overset{\text{def}}{=} \Delta\alpha\right) \quad \text{for all } i$$

(More specifically, let $n \in \mathbb{N}$ be fixed. By the Intermediate Value Theorem, there exists $x_1 \in [a,b]$ s.t $\alpha(x_1) = \alpha(a) + \frac{\alpha(b)-\alpha(a)}{n}$. Inductively,

there exists $x_2 \in [x_1, b]$ s.t $\alpha(x_2) = \alpha(a) + 2 \cdot \frac{\alpha(b)-\alpha(a)}{n}$

$\vdots$

$x_{n-1} \in [x_{n-2}, b]$ s.t $\alpha(x_{n-1}) = \alpha(a) + (n-1) \cdot \frac{\alpha(b)-\alpha(a)}{n}$

And take $x_n = b$. Essentially, the Existence such partition is obtained by I.V.P.

But, Monotone can have only first-kind discontinuity and IVP implies it can not have first-kind discontinuity, thus Monotone IVT $\Rightarrow$ continuous)

Given $\varepsilon > 0$, wlog, Assume f is monotone increasing. Take $n \in \mathbb{N}$ s.t $\frac{[\alpha(b)-\alpha(a)]\cdot[f(b)-f(a)]}{n} < \varepsilon$.

$$U(f,P_n,\alpha) - L(f,P_n,\alpha) = \sum_{i=1}^{n}[M_i - m_i]\cdot\Delta\alpha_i = \sum_{i=1}^{n}[f(x_i)-f(x_{i-1})]\Delta\alpha_i = \Delta\alpha \cdot \sum_{i=1}^{n}[f(x_i)-f(x_{i-1})]$$

$$= \Delta\alpha \cdot [f(b)-f(a)] < \varepsilon. \qquad \blacksquare$$

Proof of Thm. 0.1.10.

Since ϕ is continuous on the compact set $[m, M]$, it is continuous uniformly.

Given $\varepsilon > 0$, there exists $0 < \delta < \varepsilon$ s.t. $|s-t| < \delta \Rightarrow |\phi(s) - \phi(t)| < \varepsilon$.

Since $f \in \mathscr{R}(\alpha)$ on $[a,b]$, there is a partition $\{x_0, \cdots, x_n\}$ on $[a,b]$ s.t.

$$\mathcal{U}(f, P, \alpha) - \mathcal{L}(f, P, \alpha) < \delta^2$$

Let $M_i = \sup f(t)$ and $m_i = \inf f(t)$ and $M_i^* = \sup (\phi \circ f)(t)$ and $m_i^* = \inf (\phi \circ f)(t)$ where the sup, inf values are taken in $t \in [x_{i-1}, x_i]$.

Divides the index set $\{1, \cdots, n\}$ as;

$$A = \{ i \in \{1, \cdots, n\} \mid M_i - m_i < \delta \} \text{ and } B = \{ i \in \{1, \cdots, n\} \mid M_i - m_i \geq \delta \}.$$

Now, if $i \in A$, then $M_i^* - m_i^* = \sup (\phi \circ f)(t) - \inf (\phi \circ f)(t) \leq \varepsilon$

because $\forall s, t \in [x_{i-1}, x_i]$, $|f(s) - f(t)| \leq M_i - m_i \Rightarrow |\phi(f(s)) - \phi(f(t))| < \varepsilon$.

And, if $i \in B$, then $M_i^* - m_i^* \leq 2 \cdot \sup\limits_{t \in [a,b]} |\phi(t)|$

We obtain that

$$\delta \cdot \sum_{i \in B} \Delta\alpha_i \leq \sum_{i \in B} [M_i - m_i] \cdot \Delta\alpha_i \leq \sum_{i=1}^{n} [M_i - m_i] \cdot \Delta\alpha_i = \mathcal{U}(f, P, \alpha) - \mathcal{L}(f, P, \alpha) < \delta^2.$$

That is $\sum_{i \in B} \Delta\alpha_i < \delta$. Finally,

$$\begin{aligned}
\sum_{i=1}^{n} [M_i^* - m_i^*] \Delta\alpha_i &= \sum_{i \in A} [M_i^* - m_i^*] \Delta\alpha_i + \sum_{i \in B} [M_i^* - m_i^*] \cdot \Delta\alpha_i \\
&\leq \varepsilon \cdot \sum_{i \in A} \Delta\alpha_i + 2 \cdot \sup_{t \in [m, M]} |\phi(t)| \cdot \sum_{i \in B} \Delta\alpha_i \\
&\leq \varepsilon \cdot [\alpha(b) - \alpha(a)] + 2 \sup |\phi| \cdot \delta \\
&\leq \varepsilon \cdot [\alpha(b) - \alpha(a) + 2 \sup |\phi|]
\end{aligned}$$

⌜Proof of Thm 0.1.11.

Suppose that $f \in \mathcal{R}(\alpha)$. Given $\varepsilon > 0$, there is a partition $P = \{x_0, \cdots, x_n\}$ on $[a,b]$ s.t.

$$\mathcal{U}(\alpha', P) - \mathcal{L}(\alpha', P) < \varepsilon,$$ since α' is Continuous, thus $\alpha' \in \mathcal{R}$.

By the Mean-Value Theorem, for each $1 \le i \le n$

$$\Delta \alpha_i = \alpha(x_i) - \alpha(x_{i-1}) = \alpha'(t_i)(x_i - x_{i-1}) = \alpha'(t_i) \cdot \Delta x_i \text{ for some } t_i \in [x_{i-1}, x_i]$$

For any $s_i \in [x_{i-1}, x_i]$,

$$\sum_{i=1}^{n} |\alpha'(s_i) - \alpha'(t_i)| \cdot \Delta x_i \le \sum_{i=1}^{n} [M_i - m_i] \cdot \Delta x_i < \varepsilon.$$

Now,

$$\left| \sum_{i=1}^{n} f(s_i) \cdot \Delta \alpha_i - \sum_{i=1}^{n} f(s_i) \cdot \alpha'(s_i) \cdot \Delta x_i \right|$$

$$= \left| \sum_{i=1}^{n} f(s_i) \cdot \alpha'(t_i) \cdot \Delta x_i - \sum_{i=1}^{n} f(s_i) \alpha'(s_i) \cdot \Delta x_i \right|$$

$$= \left| \sum_{i=1}^{n} [\alpha'(t_i) - \alpha'(s_i)] \cdot f(s_i) \cdot \Delta x_i \right| \le \sup|f| \cdot \varepsilon \quad (*)$$

Therefore, $$\left| \overline{\int_a^b} f \, d\alpha - \overline{\int_a^b} f \alpha' \, dx \right| \le \left| \mathcal{U}(f, P, \alpha) - \mathcal{U}(f\alpha', P, x) \right| \le \sup|f| \cdot \varepsilon$$

since $(*)$ holds for any refinement of P.

0.1.3 Riemann-Integral

Definition 0.1.12. Let $\mathcal{P}[a,b]$ be a set of partitions. Define a *Norm* as:

$$\|\cdot\| : \mathcal{P}[a,b] \to \mathbb{R} : P \mapsto \max\{|x_i - x_{i-1}| \mid i = 1,2,\ldots,n\} \text{ where } P = \{x_0, x_1, \ldots, x_n\}$$

Theorem 0.1.13. Let $f : [a,b] \to \mathbb{R}$ and $A \in \mathbb{R}$, TFAE:

1. f is Riemann-Integral with $\displaystyle\int_a^b f = A$.
2. For any $\varepsilon > 0$, there exists $\delta > 0$ such that $\forall P \in \mathcal{P}[a,b],\ \|P\| < \delta \implies |R(f,P) - A| < \varepsilon$.
3. For any $\varepsilon > 0$, there exists $\delta > 0$ such that $\forall P \in \mathcal{P}[a,b],\ P \supseteq P_0 \implies |R(f,P) - A| < \varepsilon$.

The statement 2) can be written as: $\displaystyle\lim_{\|P\|\to 0} R(f,P) = A$.

Proof of Thm 0.1.13.

Enough to show 1) <=> 2). Let $\int_a^b f = A$.

Given $\varepsilon > 0$, there exists $P_0 = \{x_0, x_1, \cdots, x_n\}$ s.t $U(f,P_0) < A + \varepsilon$.

Put $M = \sup|f|$, $\delta_1 = \frac{\varepsilon}{nM}$. Let $P \in \mathcal{P}[a,b]$ s.t $\|P\| < \delta_1$. Denote $P = \{y_0, \cdots, y_m\}$.

Let $I = \{i \in \{1, \cdots, m\} \mid (y_{i-1}, y_i) \cap P_0 \neq \emptyset\}$ (WLOG, $f > 0$. ($f = f + \min f$))

$J = \{j \in \{1, \cdots, m\} \mid (y_{j-1}, y_j) \cap P_0 = \emptyset\}$

Now, $U(f,P) = \sum_{i=1}^{m} M_i (y_i - y_{i-1})$

$$= \sum_{i \in I} M_i (y_i - y_{i-1}) + \sum_{i \in J} M_i (y_i - y_{i-1})$$

$$< nM\|P\| + U(f,P_0) < A + 2\varepsilon.$$

겹치는게 많아봐야 n개! 더 큰 구간에서 sup!

Similarly $L(f,P) > A - 2\varepsilon$, for some $\delta_2 > 0$, $\|P\| < \delta_2$

Theorem 0.1.14. Let $f : [a, b] \to \mathbb{R}$ be a Riemann-Integrable function. Define

$$F : [a, b] \to \mathbb{R} : x \mapsto \int_a^x f(t)dt$$

Then, F is Continuous on $[a, b]$. Particular, if f is continuoous at $x_0 \in [a, b]$, then F is differentiable at x_0 with

$$F'(x_0) = f(x_0)$$

Theorem 0.1.15. The fundamental theorem of calculus
If $f : [a, b] \to \mathbb{R}$ is Riemann-Integrable and there exists differetiable function $F : [a, b] \to \mathbb{R}$ such that $F' = f$, then

$$\int_a^b f(x)dx = F(b) - F(a)$$

0.1.4 Bounded Variation Function

Definition 0.1.16. Let $f : [a, b] \to \mathbb{R}$ be a bounded function and $P = \{x_0, x_1, \ldots, x_n\} \in \mathcal{P}[a, b]$.
Define a *Variation* of f with respects to P as:

$$V_a^b(f, P) \overset{\text{def}}{=} \sum_{i=1}^{n} |f(x_i) - f(x_{i-1})|$$

If the set $\{V_a^b(f, P) \mid P \in \mathcal{P}[a, b]\}$ is bounded above, then f is called *Bounded Variation Function*.
In this case, define a *Total Variation* of f as:

$$TV_a^b(f) \overset{\text{def}}{=} \sup\{V_a^b(f, P) \mid P \in \mathcal{P}[a, b]\}$$

In simply, denote $V(f, P) = V_a^b(f, P)$ and $TV(f) = TV_a^b(f)$.

Lemma 0.1.17. Let $f, g : [a, b] \to \mathbb{R}$ be Bounded Variation Functions, $\alpha \in \mathbb{R}$ be a real number.
Then, $f + g, fg, \alpha f$ are Bounded Variation Functions and satisfy

$$\begin{aligned}
&TV(f+g) \leq TV(f) + TV(g) \\
&TV(fg) \leq \sup_{t\in[a,b]} |g(t)| \cdot TV(f) + \sup_{t\in[a,b]} |f(t)| \cdot TV(g) \\
&TV(\alpha f) = |\alpha| \cdot TV(f)
\end{aligned}$$

Lemma 0.1.18. Let $f : [a, b] \to \mathbb{R}$ be a map, and $a < c < b$.

f is bounded variation on $[a, b]$ *if and only if* f is bounded variation on $[a, c]$ and $[c, b]$

In this case, the following equality holds:

$$TV_a^c(f) + TV_c^b(f) = TV_a^b(f)$$

Theorem 0.1.19. Let $f : [a, b] \to \mathbb{R}$ be given.

f is bouned varation *if and only if* For some monotone increasing $g, h : [a, b] \to \mathbb{R}$, $f = g - h$

Proof. Suppose f is bounded variation. Define

$$g(x) = V_a^x(f) \quad \text{and} \quad h(x) = V_a^x(f) - f(x).$$

By definition, g is monotone increasing. And, if $a \leq x < y \leq b$, then

$$h(y) - h(x) = V_a^y(f) - V_a^x(f) - (f(y) - f(x)) = V_x^y(f) - (f(y) - f(x)) \geq 0.$$

The converse is trivial. $\blacksquare$

Proposition 0.1.20. Let $f : [a, b] \to \mathbb{R}$ be a continuously differentiable function.
If f is bounded variation, then

$$\int_a^b |f'(x)|dx \leq TV_a^b(f)$$

Proof. Let $P = \{x_0, \ldots, x_n\}$ be a Partition on $[a, b]$. By the Mean-Value Theorem, for each $1 \leq i \leq n$, for some $t_i \in [x_{i-1}, x_i]$,

$$V_a^b(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i$$

Meanwhile,

$$L(|f'|, P) \leq \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i \leq U(|f'|, P)$$

Both Lower Sum and Upper Sum converge to the $\displaystyle\int_a^b |f'(x)|dx$, we can complete this conversation.
Given $\varepsilon > 0$, there exists a partition P on $[a, b]$ such that

$$\int_a^b |f'(x)|dx \leq V_a^b(f, P) + \varepsilon \leq TV_a^b(f) + \varepsilon$$

Since $\varepsilon > 0$ was arbitrary, therefore we obtain the result. □

Note: if f is continuously differentiable, then

For any $\varepsilon > 0$, there exists a partition P on $[a, b]$ such that $\left| \displaystyle\int_a^b |f'(x)|dx - V_a^b(f, P) \right| < \varepsilon$.

And, we can weaken the condition: continuously differentiable $\to$ differentiable and the derivative is Riemann-Integrable.

Proposition 0.1.21. If $f : [a, b] \to \mathbb{R}$ is differentiable and the derivative f' is bounded,
then f is bounded variation.

Proof. For any partition P on $[a, b]$, by the Mean Value Theorem, there exists $t_i \in (x_{i-1}, x_i)$ such that

$$V(f, P) = \sum_{i=1}^n |f(x_i) - f(x_{i-1})| = \sum_{i=1}^n |f'(t_i)| \cdot \Delta x_i \leq \sum_{i=1}^n \sup |f'| \cdot \Delta x_i = \sup |f'| \cdot 1$$

Since P was arbitrary, f is bounded variation. □

Definition 0.1.22. Let $\alpha : [a, b] \to \mathbb{R}$ be a bounded variation.
A map f is said to be *Riemann-Stieltjes integrable* with respect to α if:

for some monotone increasing α_1, α_2, $\alpha = \alpha_1 - \alpha_2$ such that $f \in \mathcal{R}(\alpha_1)$ and $f \in \mathcal{R}(\alpha_2)$

and in this case, define

$$\int_a^b f d\alpha = \int_a^b f d\alpha_1 - \int_a^b f d\alpha_2$$

Lemma 0.1.23. If $f : [a, b] \to \mathbb{R}$ is continuous, then $f \in \mathcal{R}(\alpha)$ for any bounded varation α.

Theorem 0.1.24. Let $f, \alpha : [a, b] \to \mathbb{R}$ be bounded variation functions. If $f \in \mathcal{R}(\alpha)$, then $\alpha \in \mathcal{R}(f)$ with

$$\int_a^b f d\alpha + \int_a^b \alpha df = f(b)\alpha(b) - f(a)\alpha(a)$$

Corollary 0.1.25. If $f : [a, b] \to \mathbb{R}$ is monotone increasing and $g : [a, b] \to \mathbb{R}$ is continuous.
Then, there exists $c \in [a, b]$ such that

$$\int_a^b f(x)g(x)dx = f(a)\int_a^c g(x)dx + f(b)\int_c^b g(x)dx$$

0.1.5 Step function

Definition 0.1.26. Define a *Step function* as:

$$I : \mathbb{R} \to \mathbb{R} : x \mapsto \begin{cases} 0 & \text{if } x \leq 0 \\ 1 & \text{if } x > 0 \end{cases}$$

Lemma 0.1.27. If $a < s < b$, f is bounded map on $[a, b]$, f is continuous at s, and $\alpha(x) = I(x - s)$, then

$$\int_a^b f d\alpha = f(s)$$

Theorem 0.1.28. Suppose that $c_n \geq 0$ and $\sum c_n$ converges. And, $\{s_n\}$ is a sequence of distinct points in (a, b). Define

$$\alpha(x) = \sum_{n=1}^{\infty} c_n I(x - s_n)$$

If f is Continuous on $[a, b]$, then

$$\int_a^b f d\alpha = \sum_{n=1}^{\infty} c_n f(s_n)$$